

Hyakumanben Intersection Traffic Analysis Exercise

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1. Data Collection

Traffic data for this exercise were collected by Wang, Shen and Song at the Hyakumanben intersection. The field survey was conducted on December 22, 2025, from approximately 3:20 p.m. to 4:10 p.m. Vehicle counts were recorded for the East-West, West-South, North-East, and South-East movements. Pedestrian volumes were also observed at the four crosswalks on the east, west, north, and south sides of the intersection. The observed counts were converted into hourly traffic and pedestrian volumes using appropriate expansion factors. Additional hourly traffic volume data were obtained by referencing the survey results from other groups.

In addition, the signal cycle time and green times for each phase were measured over multiple consecutive cycles to ensure data reliability. During the survey, Song was responsible for vehicle counting, Wang recorded pedestrian volumes at the north and east crosswalks, and Shen recorded pedestrian volumes at the south and west crosswalks. The observation positions of each surveyor are illustrated in Fig. 1.



Fig. 1: Observation positions of surveyors (red: Song; blue: Wang; orange: Shen)

2. Data Summary and Analysis

2.1 Traffic Signal Settings

Based on field observations, the traffic signals at the intersection follow a consistent operational sequence. For either the east-west or north-south direction, through movements, turning movements, and pedestrian crossings are initially given green signals simultaneously. The pedestrian signal then turns red first, followed by the through movement signal approximately 10 seconds later. After an additional 7 seconds, the turning movement signal turns red. This

sequence is followed by an all-red interval of approximately 3 seconds, after which the cycle repeats.

It was observed that the total time alternates between the east-west and north-south directions across consecutive cycles. For example, in one cycle, the east-west direction has a total time of 68 seconds while the north-south direction has 62 seconds, whereas in the following cycle, the east-west direction has 62 seconds, and the north-south direction has 68 seconds. This alternating pattern continues throughout the observation period. Therefore, average values were adopted as representative signal timings.

Based on these averaged values, the signal control diagram shown in Fig. 2 was developed. The pedestrian green time is 45 seconds, the through movement green time is 55 seconds, and the turning movement green time is 62 seconds. This timing structure applies to both the east-west and north-south directions, resulting in a total signal cycle time of 130 seconds.

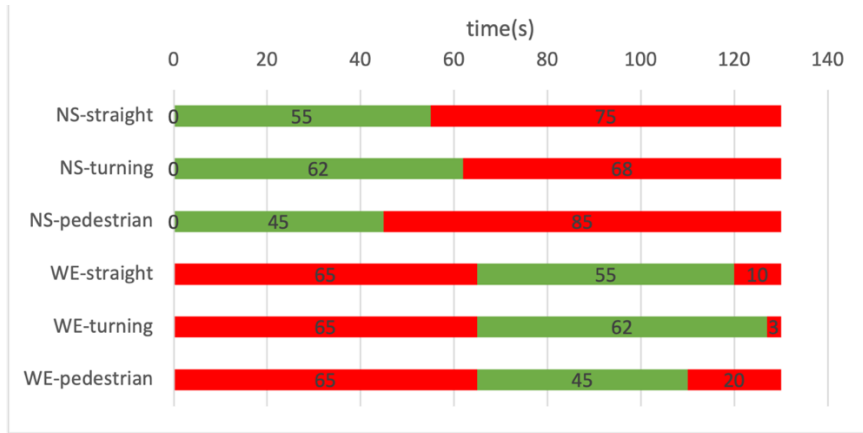


Fig. 2: Signal Control Diagram

2.2 Traffic Flow Analysis

The observed hourly vehicle and pedestrian volumes are summarized in Table 1, together with calculated performance-related parameters:

Table 1: Intersection Performance under Field Survey Conditions

	α	turning factor	a_eq	β	ρ	$q=w\alpha$	$q=p+0.5\rho^2p/(1-p)$	ρ^2s	w_u	w_r	w	movement	g	r	q_max
NS	440	1	440	3000	0.146667	2.78886	0.438639456	0.346667	25.35306	0.000209	22.81795	straight	55	75	9.166667
NE	90	1.39	125.1	1500	0.0834	0.606828	0.193401315	0.174871	19.40281	0.000148	17.46266	left turn	62	68	2.363
NW	125	3.63	453.75	1500	0.3025	2.892528	1.184281529	0.634274	25.49766	0.001212	22.94898	right turn	62	68	8.570833
SN	496	1	496	3000	0.165333	3.21412	0.516126132	0.390788	25.92007	0.000253	23.32829	straight	55	75	10.33333
SW	140	1.5	210	1500	0.14	1.085704	0.354536751	0.293548	20.67979	0.00029	18.61207	left turn	62	68	3.966667
SE	97	2.82	273.54	1500	0.18236	1.487482	0.50072708	0.382368	21.75116	0.000433	19.57643	right turn	62	68	5.166667
WE	296	1	296	3000	0.098667	1.776224	0.268676879	0.233212	24.0029	0.00012	21.60272	straight	55	75	6.166667
WN	104	1.46	151.84	1500	0.101227	0.751146	0.240843435	0.212249	19.78765	0.000188	17.80906	left turn	62	68	2.868089
WS	118	2.2	259.6	1500	0.173067	1.395811	0.466224496	0.362882	21.50671	0.000398	19.3564	right turn	62	68	4.903556
EW	270	1	270	3000	0.09	1.604773	0.241467562	0.212727	23.7743	0.000106	21.39697	straight	55	75	5.625
ES	81	1.33	107.73	1500	0.07182	0.51605	0.163939267	0.15059	19.16074	0.000124	17.24478	left turn	62	68	2.0349
EN	49	2.32	113.68	1500	0.075787	0.546889	0.173918719	0.158908	19.24298	0.000132	17.3188	right turn	62	68	2.147289
pedestrians	counting	turning factor	adjusted												
North	456	1.61	1.46												
South	312	1.43	1.33												
East	400	1.52	1.39												
West	496	1.67	1.50												

2.2.1 Definition of parameters

The arrival rate α represents the hourly traffic demand (veh/h) and was obtained by scaling the observed vehicle counts. Pedestrian volumes were similarly converted into hourly flows (peds/h).

Turning factors were applied to account for capacity reductions for turning movements caused by conflicting pedestrians and traffic streams. Although turning movements have a total green time of 62 seconds, part of this time is obstructed and cannot be fully utilized. For left turns, pedestrian crossings conflict with turning vehicles during the first 45 seconds of the green phase. The corresponding turning factor was determined based on pedestrian volumes using Table 2. During the remaining 17 seconds, no conflicting pedestrian movements exist, and the left turn is treated as a protected movement with a turning factor of 1.05. The overall left-turn turning factor was calculated as a weighted average of the obstructed and protected periods. For right turns, conflicting traffic streams exist during the first 55 seconds of the green phase. Given that each approach consists of two through lanes, one left-turn lane, and one right-turn lane, the obstructed turning factor was determined using the middle column of Table 3. During the remaining 7 seconds, the right turn is considered protected and assigned a turning factor of 1.05. The overall right-turn turning factor was calculated as a weighted average in the same manner. Since Tables 2 and 3 provide turning factors only for discrete traffic volumes, linear interpolation between adjacent values was used to determine the turning factors for the observed volumes. For through movements, turning factor equals 1.

Table 2: Through-vehicle equivalents for left turning vehicles

Pedestrian volume in conflicting crosswalks (peds/h)	Equivalent
	1
None (0)	1.18
Low (50)	1.21
Moderate (200)	1.32
High (400)	1.52
Very High (800)	2.14

Table 3: Through-vehicle equivalents for right turning vehicles

Opposing flow (veh/h)	Number of opposing lanes		
	1	2	3
0	1.1	1.1	1.1
200	2.5	2	1.8
400	5	3	2.5
600	10	5	4
800	13	8	6
1000	15	13	10
>1200	15	15	15

α_{eq} is the equivalent arrival rate. $\alpha_{eq} = \alpha \times \text{turning factor}$.

β is the maximum departure rate (capacity). It was determined based on the value of 1500 veh/lane/gh provided in the lecture materials. For the Hyakumanben intersection, each approach consists of two through lanes, one left-turn lane, and one right-turn lane. Therefore, the capacity for through movements was set to 3000 veh/gh, while the capacity for turning movements was set to 1500 veh/gh.

ρ is the degree of saturation. $\rho = \alpha/\beta$.

ρ_s is the effective degree of saturation. It was further calculated to account for the proportion of green time within cycle time. $\rho_s = \alpha c/\beta g$.

q is the expected queue length. It was estimated using two different methods. In the first method $q = w \times \alpha$. The second method applied the Pollaczek-Khintchine formula by modeling the intersection as a system with random arrivals and uniform service, and $q = \rho + 0.5 \rho^2/(1 - \rho)$. The first method considers the total time a vehicle spends in the system, including waiting during the red signal and queuing during the green phase before passing the stop line, whereas the second method considers only the queue formed during the red signal. As a result, the first method yields a larger expected queue and is considered more appropriate for this analysis.

g and r denote the green time and red time for each movement within a signal cycle.

w is the expected delay. It was calculated by combining uniform delay w_u and random delay w_r , both of which were obtained using the Webster formula, as shown below. In the calculations, c is the cycle time 130s.

$$\begin{aligned} w_u &= 0.5 r^2 / c(1 - \alpha/\beta) \\ w_r &= \rho_s^2 / 2\alpha(1 - \rho_s) \\ w &= 0.9(w_u + w_r) \end{aligned}$$

q_{max} is the maximum expected queue length within a cycle. $q_{max} = r \cdot \alpha/3600$.

2.2.2 Result analysis

Based on the results presented in Table 1, the Hyakumanben intersection was operating under unsaturated conditions during the survey period. No abnormal values of degree of saturation exceeding 1 were observed, and all movements exhibited saturation levels well below their capacity limits. Delays and queues remained moderate, with no evidence of spillback, indicating stable system operation.

The primary performance constraints were found to originate from signal timing and turning conflicts rather than insufficient capacity. Traffic demand in the north-south direction was generally higher than that in the east-west direction, suggesting that equal green time allocation for both directions may not be optimal. In addition, turning movements were significantly affected by pedestrian crossings and conflicting traffic streams. Relatively large turning factors resulted in higher equivalent arrival rates, which could potentially lead to operational issues during peak periods.

2.2.3 Peak period analysis

Table 4 summarizes the peak hour conditions at the Hyakumanben intersection. Peak hour demand was represented by multiplying each arrival rate by 1.5, while all other calculation procedures remained unchanged. Under peak-hour conditions, the intersection exhibits signs of over-saturation, primarily associated with right-turn movements. In particular, values of the effective degree of saturation ρ_s exceed 1, accompanied by substantial increases in expected queue length and maximum queue length, indicating a reduction in overall system stability.

The most critical abnormalities occur at the NW and SE right-turn movements. These issues are mainly caused by large turning factors, which result in excessively high equivalent arrival rates α_{eq} . Therefore, the primary improvement priority is to separate turning movements from through movements and treat them as fully protected turns, in order to mitigate turning conflicts and restore stable operation under peak-hour demand.

Table 4: Intersection Performance under Peak Hour Conditions

peak hour	α	turning factor	a_eq	β	ρ	$q=w\alpha$	$q=p+0.5p\rho/(1-p)$	ρ^2s	w_u	w_r	w	movement	g	r	q_max
NS	660	1	660	3000	0.22	4.576624	0.801666667	0.52	27.73669	0.000427	24.9634	straight	55	75	13.75
NE	135	1.62	218.7	1500	0.1458	1.138361	0.373014668	0.30571	20.8202	0.000308	18.73846	left turn	62	68	4.131
NW	187.5	6.47	1213.125	1500	0.80875	28.20205	-0.370749216	1.695766	92.99145	-0.0017	83.69077	right turn	62	68	22.91458
SN	744	1	744	3000	0.248	5.351219	1.001351015	0.586182	28.76944	0.000558	25.89299	straight	55	75	15.5
SW	210	1.78	373.8	1500	0.2492	2.213673	0.808413854	0.522516	23.68755	0.000765	21.31949	left turn	62	68	7.060667
SE	145.5	5.35	778.425	1500	0.51895	7.192993	-5.629958002	1.088121	36.97041	-0.00863	33.2656	right turn	62	68	14.70358
WE	444	1	444	3000	0.148	2.818618	0.443924751	0.349818	25.39274	0.000212	22.85366	straight	55	75	9.25
WN	156	1.71	266.76	1500	0.17784	1.442637	0.483753859	0.37289	21.63157	0.000416	19.46879	left turn	62	68	5.0388
WS	177	2.82	499.14	1500	0.33276	3.326221	1.502970952	0.697723	26.654	0.001613	23.99005	right turn	62	68	9.4282
EW	405	1	405	3000	0.135	2.532394	0.393857871	0.319091	25.01112	0.000185	22.51017	straight	55	75	8.4375
ES	121.5	1.47	178.605	1500	0.11907	0.90145	0.291198611	0.249663	20.18845	0.000233	18.16982	left turn	62	68	3.37365
EN	73.5	3.17	232.995	1500	0.15533	1.226453	0.404346813	0.325692	21.0551	0.000338	18.9499	right turn	62	68	4.401017
pedestrians	counting	turning factor	adjusted												
North	684	1.96	1.71												
South	468	1.63	1.47												
East	600	1.83	1.62												
West	744	2.05	1.78												

3. Proposed Changes

3.1 Revised Traffic Signal Settings

As discussed earlier, the signal optimization focuses on two key aspects. Firstly, all turning movements are separated from through movements and operated as fully protected turns, in order to reduce the turning factor and consequently lower the equivalent arrival rate α_{eq} . Secondly, the green time allocation between the north-south (NS) and west-east (WE) directions is rebalanced, while ensuring sufficient crossing time for pedestrians.

Based on Google Earth satellite images, the pedestrian crossing length at the Hyakumanben intersection is approximately 20 m. Therefore, pedestrian green time should be at least 30s to ensure safe crossing. Taking these considerations into account, the green splits of different signal phases are redesigned. The cycle time remains unchanged at 130 s. Given the higher traffic demand in the NS direction, 70 s is allocated to NS movements and 60 s to WE movements.

For the NS direction, the first 40 s are assigned to through traffic together with the corresponding pedestrian green phase. This is followed by 27 s of protected turning movements, including both left-turn and right-turn traffic, and finally 3 s of all-red time. For the WE direction, the first 37 s are allocated to through movements and pedestrians, followed by 20 s of protected left-turn and right-turn movements, and 3 s of all-red time. The detailed signal control diagram is shown in Fig. 3.

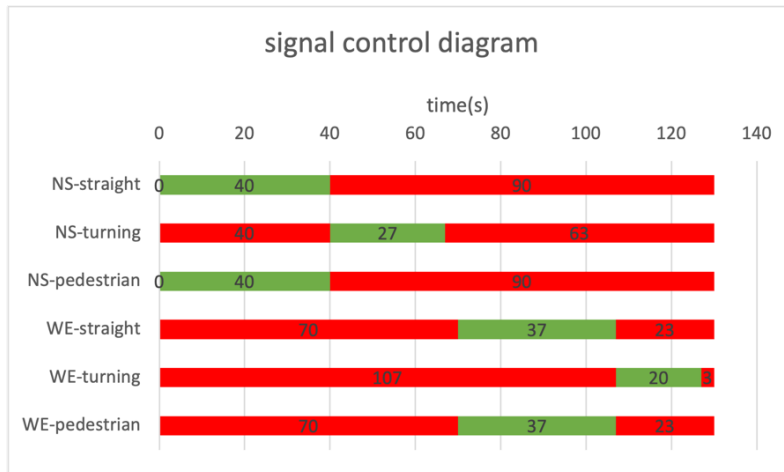


Fig. 3: Revised signal control diagram

Under this scheme, although the green time for turning movements is significantly shorter, traffic operations during the turning phase become smoother due to the absence of conflicting vehicle and pedestrian flows. However, this adjustment reduces the green time available for through movements, resulting in higher saturation ratios for straight-through traffic. Overall, the proposed signal plan provides a more coordinated and balanced green time distribution, avoiding abnormal saturation rates and improving the overall operational stability of the intersection.

3.2 Re-evaluated Traffic Performance

Tables 5 and 6 present the intersection performance under the revised traffic signal settings using field survey traffic volumes and estimated peak-hour demand, respectively. The results indicate that under peak-hour conditions, the intersection no longer exhibits oversaturated behavior. The maximum value of ρ^s is around 0.8, with no abnormal cases exceeding 1. Queue lengths become more balanced and controllable. In particular, the maximum expected queue length is concentrated in the range of approximately 4 to 11 vehicles, which is consistent with realistic operating conditions and unlikely to spill back to adjacent intersections.

As discussed in Section 3.1, these improvements are achieved at the expense of other performance aspects. The expected queue length and expected delay for through movements increase compared to the original signal settings, but they remain within a reasonable range. Meanwhile, the expected queue length and delay across different phases become more evenly distributed, avoiding the extreme imbalance observed before the modification, where one phase experienced excessively large values while others remained relatively small. Overall, the improvement sacrifices marginal efficiency to ensure operational stability and robustness under peak-hour conditions and is therefore considered feasible for implementation.

Table 5: Intersection Performance under Field Survey Conditions (revised signals)

	α	turning factor	a_eq	β	ρ	$q=w\alpha$	$q=p+0.5^*p^s/(1-p)$	ρ^s	w.u	w.r	w	movement	g	r	q_max
NS	440	1	440	3000	0.1466667	4.0159798	0.693747346	0.4766667	36.508413	0.0004934	32.858016	straight	40	90	11
NE	90	1.05	94.5	1500	0.063	1.0288221	0.369370016	0.3033333	43.547328	0.0006988	39.193224	left turn	27	103	2.70375
NW	125	1.05	131.25	1500	0.0875	1.4672999	0.574648148	0.4212963	44.716544	0.0011684	40.245941	right turn	27	103	3.7552083
SN	496	1	496	3000	0.1653333	4.6283651	0.849358309	0.5373333	37.324896	0.0006291	33.592972	straight	40	90	12.4
SW	140	1.05	147	1500	0.098	1.6625154	0.682629993	0.4718519	45.23708	0.0014339	40.714662	left turn	27	103	4.2058333
SE	97	1.05	101.85	1500	0.0679	1.1146727	0.40632325	0.3269259	43.776254	0.0007796	39.39933	right turn	27	103	2.9140417
WE	296	1	296	3000	0.0986667	2.7311308	0.438639456	0.3466667	36.906862	0.0003107	33.216455	straight	37	93	7.6466667
WN	104	1.05	109.2	1500	0.0728	1.3703077	0.685726803	0.4732	50.192474	0.0019462	45.174978	left turn	20	110	3.3366667
WS	118	1.05	123.9	1500	0.0826	1.5713977	0.848130415	0.5369	50.728648	0.0025119	45.658044	right turn	20	110	3.7858333
EW	270	1	270	3000	0.09	2.4675056	0.389333405	0.3162162	36.555368	0.0002708	32.900075	straight	37	93	6.975
ES	81	1.05	85.05	1500	0.0567	1.0490294	0.476103332	0.36855	49.335801	0.0012646	44.403359	left turn	20	110	2.59875
EN	49	1.05	51.45	1500	0.0343	0.6198702	0.254934237	0.22295	48.191428	0.0006217	43.372844	right turn	20	110	1.5720833
pedestrians	counting	turning factor													
North	456	1.05													
South	312	1.05													
East	400	1.05													
West	496	1.05													

Table 6: Intersection Performance under Peak-Hour Conditions (revised signals)

peak hour	α	turning factor	a_eq	β	ρ	$q=w\alpha$	$q=p+0.5^*p^s/(1-p)$	ρ^s	w.u	w.r	w	movement	g	r	q_max
NS	660	1	660	3000	0.22	6.5904609	1.611885965	0.715	39.940828	0.0013589	35.947969	straight	40	90	16.5
NE	135	1.05	141.75	1500	0.0945	1.5969401	0.644931193	0.455	45.062227	0.0013399	40.55721	left turn	27	103	4.055625
NW	187.5	1.05	196.875	1500	0.13125	2.3118643	1.174462788	0.6319444	46.968456	0.0027556	42.27409	right turn	27	103	5.6328125
SN	744	1	744	3000	0.248	7.7060241	2.480319588	0.806	41.427987	0.0022504	37.287214	straight	40	90	18.6
SW	210	1.05	220.5	1500	0.147	2.6371569	1.564915505	0.7077778	47.835693	0.0038872	43.055622	left turn	27	103	6.30875
SE	145.5	1.05	152.775	1500	0.10185	1.735239	0.726334748	0.4903889	45.430993	0.0015444	40.889283	right turn	27	103	4.3710625
WE	444	1	444	3000	0.148	4.3339409	0.801666667	0.52	39.043879	0.0006344	35.140062	straight	37	93	11.47
WN	156	1.05	163.8	1500	0.1092	2.1395861	1.57784969	0.7098	52.243446	0.0052994	47.023871	left turn	20	110	5.005
WS	177	1.05	185.85	1500	0.1239	2.4685061	2.471388075	0.80535	53.120034	0.0089644	47.816098	right turn	20	110	5.67875
EW	405	1	405	3000	0.135	3.8938341	0.688318975	0.4743243	38.457092	0.0005284	34.611858	straight	37	93	10.4625
ES	121.5	1.05	127.575	1500	0.08505	1.6223446	0.894542986	0.552825	50.864486	0.0026786	45.780448	left turn	20	110	3.898125
EN	73.5	1.05	77.175	1500	0.05145	0.9466252	0.418442639	0.334425	49.062739	0.0010887	44.157445	right turn	20	110	2.358125
pedestrians	counting	turning factor													
North	684	1.05													
South	468	1.05													
East	600	1.05													
West	744	1.05													

4. SUMO Simulation and Comparison with Theoretical Results

To validate the analytical results and evaluate the operational performance of the Hyakumanben intersection under different demand and signal control settings, microscopic traffic simulations were conducted using Simulation of Urban MObility (SUMO). Four simulation scenarios were prepared to match the analytical cases presented in Sections 2 and 3: Scenario 1 (Original – Off-peak), Scenario 2 (Original – Peak), Scenario 3 (Revised – Off-peak), and Scenario 4 (Revised – Peak). Two types of outputs were used: (i) trip-based outputs (`tripinfo.xml`) to compute network-level experienced delay and waiting time, and (ii) E2 lane-area detector outputs to compute lane-level queues, arrival rates, and saturation levels. The capacity was defined as 1500 veh/lane/green-hour and converted into effective hourly capacity using the green ratio g/C for each movement.

4.1 Simulation setup

The SUMO network replicates the geometry of the Hyakumanben intersection, including approach lane configurations and turning movements. Fixed-time signal plans were implemented according to the original and revised timing diagrams. Peak-hour scenarios were generated by scaling the observed arrival rates by a factor of 1.5, following the analytical setup.

4.2 Performance metrics

Network-level delay was measured using `timeloss` from `tripinfo` output. Lane-level operational indicators (queues and saturation) were computed from E2 detector statistics. For each lane, the degree of saturation was calculated as $\rho = q/\beta$, where q is the simulated arrival flow (veh/h) and $\beta = 1500 \cdot (g/C)$ is the effective capacity per lane (veh/h).

4.3 Results and comparison

Overall, simulation results are consistent with analytical expectations: peak-hour scenarios produce higher delays and longer queues, while the revised signal plan improves stability by balancing phase allocation and providing protected turning phases. A notable discrepancy is that $\rho > 1$ occurs only on one of the two NS-through lanes, while the other parallel through lane remains below saturation. This indicates a lane-use imbalance in SUMO (traffic assignment and lane-changing dynamics) rather than full approach-wide oversaturation; therefore, the NS through movement is not considered saturated at the aggregate level.

Table 7. Summary of SUMO simulation results (Tripinfo + E2 indicators)

Part A — Network-level performance

Scenario	Completed trips (N)	Avg delay (s/veh)	Max delay (s)	Avg waiting (s/veh)	Max waiting (s)	Avg travel time (s/veh)	Avg speed (m/s)	Avg total delay (s/veh)
1	2313	46.01	571.19	35.93	525.00	56.32	4.91	73.65
2	3462	48.07	214.13	38.24	194.00	58.38	4.21	368.18
3	2313	45.60	119.65	36.12	11.00	55.90	3.96	47.51
4	3462	48.25	120.76	38.39	109.00	58.57	4.16	361.50

Part B — Intersection lane-level performance

Scenario	Mean ρ	Max ρ	Lane with max ρ	Mean queue (veh)	Max queue (veh)
1	0.28	0.79	S_Thru_1_E2	1.17	7
2	0.30	0.83	S_Thru_1_E2	1.43	5
3	0.49	1.08	S_Thru_1_E2	1.17	7
4	0.55	1.12	S_Thru_1_E2	1.45	5

Notes: Capacity per lane is computed using $\beta = 1500 \cdot (g/C)$ veh/h, where g is the effective green time for the lane's movement group and C is the cycle length (130 s). Saturation is computed as $\rho = q/\beta$ with q derived from simulated arrivals (veh/h).

4.4 Interpretation of simulation results (SUMO vs. theory)

The SUMO simulation results provide a consistent validation of the analytical findings and reveal additional lane-level operational details that cannot be captured by the simplified theoretical model. At the network level (Tripinfo output), the average time loss and waiting time remain within the same order of magnitude across all scenarios, indicating that the intersection operates in a generally stable regime without large-scale gridlock. However, peak-demand scenarios (Scenario 2 and Scenario 4) show substantially higher system-level delay when entry/insertion delay (departDelay) is included, which implies that the limiting factor under high demand is not only the intersection discharge process but also the limited ability of vehicles to enter the network smoothly, leading to upstream queueing effects.

From the intersection lane-level perspective (E2 detectors), the revised signal plan improves operational robustness by redistributing green time more efficiently across movements and by reducing the risk of persistent queue accumulation in turning streams. While the mean saturation degree remains below 1 in most lanes, isolated oversaturation ($\rho > 1$) appears on one of the two north-south through lanes. Importantly, the parallel lane remains unsaturated, implying that the approach is not truly oversaturated. This discrepancy is mainly attributable to SUMO's microscopic lane utilization and assignment behavior: despite identical lane capacities in theory, vehicles may not distribute evenly across parallel through lanes due to lane-changing dynamics, turning preparation, and interaction effects. Therefore, the observed $\rho > 1$ should be interpreted as a localized lane imbalance rather than a movement-wide capacity deficiency. Overall, simulation results support the analytical expectation that the revised plan alleviates critical bottlenecks under peak demand and yields a more balanced queue distribution across movements.

5. Conclusion

This study evaluated the operational performance of the Hyakumanben intersection through a combined analytical and simulation-based approach. Using traffic flow theory and fixed-time signal analysis, key performance indicators—delay, queue length, and degree of saturation—were estimated for both off-peak and peak demand conditions under the original and revised signal plans. To validate these findings and capture realistic traffic dynamics, SUMO simulations were conducted for four scenarios (original/revised $\times$ off-peak/peak), with intersection performance measured using vehicle-level trip statistics and lane-level detector outputs.

The results show that peak-hour demand significantly increases delay and queueing pressure, making the original signal plan more vulnerable to unstable operations. The revised plan improves robustness by reallocating green time and providing more structured movement control, resulting in more balanced queue formation and reduced oversaturation risks in critical movements. Although lane-level saturation values occasionally exceed 1 in simulation, this occurs only on a single north–south through lane and is likely caused by uneven lane usage inherent to microscopic simulation rather than true approach-level oversaturation. In conclusion, the revised signal plan represents an operational improvement, particularly under peak demand, and the combined analytical–simulation framework proves effective for diagnosing both theoretical capacity limitations and realistic lane-level effects.

References

Class materials – Lectures 9,10,11

SUMO Documentation – Detector Outputs, Signal Control

Highway Capacity Manual (HCM) – Signalized Intersections